MA1522 Linear Algebra for Computing
L2 Lesson 17



Tan Kai Meng

Semester 1, 2026/27

Question 1

PollEv.com/kaimengtan612

Let $\boldsymbol{A}$ be a square matrix of order n, with distinct eigenvalues $\lambda, \mu \in \mathbb{R}$. Let $S \subseteq E_\lambda$ and $T \subseteq E_\mu$. Which of the following statements is/are true?

a. If both S and T are linearly independent, then $S \cup T$ is linearly independent. [Pg 28 of Chapter 6]

b. If S and T are orthogonal, then $S \cup T$ is orthogonal.

c. If $\boldsymbol{A}$ is symmetric and both S and T are orthonormal, then $S \cup T$ is orthonormal. [Pg 55 of Chapter 6]

Question 1

PollEv.com/kaimengtan612

Let $\boldsymbol{A}$ be a square matrix of order n, with distinct eigenvalues $\lambda, \mu \in \mathbb{R}$. Let $S \subseteq E_\lambda$ and $T \subseteq E_\mu$. Which of the following statements is/are true?

a. If both S and T are linearly independent, then $S \cup T$ is linearly independent. [Pg 28 of Chapter 6] ..... **Ans: True**

b. If S and T are orthogonal, then $S \cup T$ is orthogonal.

c. If $\boldsymbol{A}$ is symmetric and both S and T are orthonormal, then $S \cup T$ is orthonormal. [Pg 55 of Chapter 6]



Question 1

PollEv.com/kaimengtan612

Let $\boldsymbol{A}$ be a square matrix of order n, with distinct eigenvalues $\lambda, \mu \in \mathbb{R}$. Let $S \subseteq E_\lambda$ and $T \subseteq E_\mu$. Which of the following statements is/are true?

a. If both S and T are linearly independent, then $S \cup T$ is linearly independent. [Pg 28 of Chapter 6] ..... **Ans: True**

b. If S and T are orthogonal, then $S \cup T$ is orthogonal. ..... **Ans: False**

c. If $\boldsymbol{A}$ is symmetric and both S and T are orthonormal, then $S \cup T$ is orthonormal. [Pg 55 of Chapter 6]



Question 1

PollEv.com/kaimengtan612

Let $\boldsymbol{A}$ be a square matrix of order n, with distinct eigenvalues $\lambda, \mu \in \mathbb{R}$. Let $S \subseteq E_\lambda$ and $T \subseteq E_\mu$. Which of the following statements is/are true?

a. If both S and T are linearly independent, then $S \cup T$ is linearly independent. [Pg 28 of Chapter 6] ..... **Ans: True**

b. If S and T are orthogonal, then $S \cup T$ is orthogonal. ..... **Ans: False**

c. If $\boldsymbol{A}$ is symmetric and both S and T are orthonormal, then $S \cup T$ is orthonormal. [Pg 55 of Chapter 6] ..... **Ans: True**



Question 1: Solution

a. If both S and T are linearly independent, then $S \cup T$ is linearly independent.

   True by Slide 28 (and 94) of Chapter 6.

b. If S and T are orthogonal, then $S \cup T$ is orthogonal.

   Let $\boldsymbol{A} = \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix}$, $S = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$, $T = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$. Then $\boldsymbol{A} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\boldsymbol{A} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$, so that $S \subseteq E_1$ and $T \subseteq E_{-1}$, and are trivially orthogonal. But $S \cup T$ is not orthogonal.

c. If $\boldsymbol{A}$ is symmetric and both S and T are orthonormal, then $S \cup T$ is orthonormal.

   By Slide 55 of Chapter 6, $\boldsymbol{u} \cdot \boldsymbol{v} = 0$ for all $\boldsymbol{u} \in S$ and $\boldsymbol{v} \in T$. Since S and T are orthonormal, therefore so is $S \cup T$.

Question 1: Solution

a. If both S and T are linearly independent, then $S \cup T$ is linearly independent.

True by Slide 28 (and 94) of Chapter 6.

b. If S and T are orthogonal, then $S \cup T$ is orthogonal.

Let $\boldsymbol{A} = \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix}$, $S = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$, $T = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$. Then $\boldsymbol{A} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\boldsymbol{A} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$, so that $S \subseteq E_1$ and $T \subseteq E_{-1}$, and are trivially orthogonal. But $S \cup T$ is not orthogonal.

c. If A is symmetric and both S and T are orthonormal, then $S \cup T$ is orthonormal.

By Slide 55 of Chapter 6, $\boldsymbol{u} \cdot \boldsymbol{v} = 0$ for all $\boldsymbol{u} \in S$ and $\boldsymbol{v} \in T$. Since S and T are orthonormal, therefore so is $S \cup T$.

Question 1: Solution

a. If both S and T are linearly independent, then $S \cup T$ is linearly independent.

True by Slide 28 (and 94) of Chapter 6.

b. If S and T are orthogonal, then $S \cup T$ is orthogonal.

Let $\boldsymbol{A} = \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix}$, $S = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$, $T = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$. Then $\boldsymbol{A} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\boldsymbol{A} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$, so that $S \subseteq E_1$ and $T \subseteq E_{-1}$, and are trivially orthogonal. But $S \cup T$ is not orthogonal.

c. If $\boldsymbol{A}$ is symmetric and both S and T are orthonormal, then $S \cup T$ is orthonormal.

By Slide 55 of Chapter 6, $\boldsymbol{u} \cdot \boldsymbol{v} = 0$ for all $\boldsymbol{u} \in S$ and $\boldsymbol{v} \in T$. Since S and T are orthonormal, therefore so is $S \cup T$.

Diagonalization

Let $\boldsymbol{A}$ be a square matrix of order n.
$\boldsymbol{A}$ is **diagonalisable** if and only if there exists an invertible matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that $\boldsymbol{A} = \boldsymbol{PDP}^{-1}$.

Let $\boldsymbol{P} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & & & d_{n-1} & 0 \\ 0 & 0 & \cdots & 0 & d_n \end{pmatrix}$. Then

$$\begin{aligned} \boldsymbol{A} = \boldsymbol{PDP}^{-1} &\Leftrightarrow \boldsymbol{AP} = \boldsymbol{PD} \\ &\Leftrightarrow \boldsymbol{A}\begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix} \begin{pmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & & & d_{n-1} & 0 \\ 0 & 0 & \cdots & 0 & d_n \end{pmatrix} \\ &\Leftrightarrow \begin{pmatrix} \boldsymbol{Av}_1 & \boldsymbol{Av}_2 & \cdots & \boldsymbol{Av}_n \end{pmatrix} = \begin{pmatrix} d_1\boldsymbol{v}_1 & d_2\boldsymbol{v}_2 & \cdots & d_n\boldsymbol{v}_n \end{pmatrix} \\ &\Leftrightarrow \boldsymbol{Av}_i = d_i\boldsymbol{v}_i \text{ for all } i. \end{aligned}$$

This shows that the columns of $\boldsymbol{P}$ are eigenvectors of $\boldsymbol{A}$ and the diagonal entries of $\boldsymbol{D}$ are the associated eigenvalues.

Diagonalization

Let $\boldsymbol{A}$ be a square matrix of order n.
$\boldsymbol{A}$ is **diagonalisable** if and only if there exists an invertible matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that $\boldsymbol{A} = \boldsymbol{PDP}^{-1}$.

Let $\boldsymbol{P} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & & & d_{n-1} & 0 \\ 0 & 0 & \cdots & 0 & d_n \end{pmatrix}$. Then

$$\begin{aligned} \boldsymbol{A} = \boldsymbol{PDP}^{-1} &\Leftrightarrow \boldsymbol{AP} = \boldsymbol{PD} \\ &\Leftrightarrow \boldsymbol{A}\begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix} \begin{pmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & & & d_{n-1} & 0 \\ 0 & 0 & \cdots & 0 & d_n \end{pmatrix} \\ &\Leftrightarrow \begin{pmatrix} \boldsymbol{Av}_1 & \boldsymbol{Av}_2 & \cdots & \boldsymbol{Av}_n \end{pmatrix} = \begin{pmatrix} d_1\boldsymbol{v}_1 & d_2\boldsymbol{v}_2 & \cdots & d_n\boldsymbol{v}_n \end{pmatrix} \\ &\Leftrightarrow \boldsymbol{Av}_i = d_i\boldsymbol{v}_i \text{ for all } i. \end{aligned}$$

This shows that the columns of P are eigenvectors of A and the diagonal entries of D are the associated eigenvalues.

Diagonalization

Let $\boldsymbol{A}$ be a square matrix of order n.
$\boldsymbol{A}$ is **diagonalisable** if and only if there exists an invertible matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that $\boldsymbol{A} = \boldsymbol{PDP}^{-1}$.

Let $\boldsymbol{P} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & & & d_{n-1} & 0 \\ 0 & 0 & \cdots & 0 & d_n \end{pmatrix}$. Then

$$\begin{aligned} \boldsymbol{A} = \boldsymbol{PDP}^{-1} &\Leftrightarrow \boldsymbol{AP} = \boldsymbol{PD} \\ &\Leftrightarrow \boldsymbol{A}\begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix} \begin{pmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & & & d_{n-1} & 0 \\ 0 & 0 & \cdots & 0 & d_n \end{pmatrix} \\ &\Leftrightarrow \begin{pmatrix} \boldsymbol{Av}_1 & \boldsymbol{Av}_2 & \cdots & \boldsymbol{Av}_n \end{pmatrix} = \begin{pmatrix} d_1\boldsymbol{v}_1 & d_2\boldsymbol{v}_2 & \cdots & d_n\boldsymbol{v}_n \end{pmatrix} \\ &\Leftrightarrow \boldsymbol{Av}_i = d_i\boldsymbol{v}_i \text{ for all } i. \end{aligned}$$

This shows that the columns of $\boldsymbol{P}$ are eigenvectors of $\boldsymbol{A}$ and the diagonal entries of $\boldsymbol{D}$ are the associated eigenvalues.

Condition for Diagonalizability

Theorem

Let $\boldsymbol{A}$ be a square matrix of order n. Then

$\boldsymbol{A}$ is diagonalisable

$\Leftrightarrow$ there exists a basis for $\mathbb{R}^n$ consisting of eigenvectors of $\boldsymbol{A}$

$\Leftrightarrow$ the roots of its characteristic polynomial are all real and the algebraic multiplicity of each eigenvalue equals its geometric multiplicity.

Remarks

Let λ be an eigenvalue of $\boldsymbol{A}$.

1. Algebraic multiplicity of $\lambda \geq$ geometric multiplicity of λ.
2. If algebraic multiplicity of λ equals 1, then geometric multiplicity of λ equals 1, so algebraic multiplicity of λ equals geometric multiplicity of λ.

Condition for Diagonalizability

Theorem

Let $\boldsymbol{A}$ be a square matrix of order n. Then

$\boldsymbol{A}$ is diagonalisable

$\Leftrightarrow$ there exists a basis for $\mathbb{R}^n$ consisting of eigenvectors of $\boldsymbol{A}$

$\Leftrightarrow$ the roots of its characteristic polynomial are all real and the algebraic multiplicity of each eigenvalue equals its geometric multiplicity.

Remarks

Let λ be an eigenvalue of $\boldsymbol{A}$.

1. Algebraic multiplicity of $\lambda \geq$ geometric multiplicity of λ.
2. If algebraic multiplicity of λ equals 1, then geometric multiplicity of λ equals 1, so algebraic multiplicity of λ equals geometric multiplicity of λ.

Condition for Diagonalizability

Theorem

Let $\boldsymbol{A}$ be a square matrix of order n. Then

$\boldsymbol{A}$ is diagonalisable

$\Leftrightarrow$ there exists a basis for $\mathbb{R}^n$ consisting of eigenvectors of $\boldsymbol{A}$

$\Leftrightarrow$ the roots of its characteristic polynomial are all real and the algebraic multiplicity of each eigenvalue equals its geometric multiplicity.

Remarks

Let λ be an eigenvalue of $\boldsymbol{A}$.

1. Algebraic multiplicity of $\lambda \geq$ geometric multiplicity of λ.
2. If algebraic multiplicity of λ equals 1, then geometric multiplicity of λ equals 1, so algebraic multiplicity of λ equals geometric multiplicity of λ.

Algorithm for Diagonalization

Let $\boldsymbol{A}$ be a square matrix of order n.

1. Find eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$ by solving $\det(x\boldsymbol{I} - \boldsymbol{A}) = 0$.

   If $\det(x\boldsymbol{I} - \boldsymbol{A})$ cannot be completely factorized in $\mathbb{R}$, then $\boldsymbol{A}$ is not diagonalizable.

2. For each (real) eigenvalue λ_i, find a basis S_i for the eigenspace E_{λ_i}.

   If $|S_i| <$ algebraic multiplicity of λ_i in $\det(\lambda\boldsymbol{I} - \boldsymbol{A})$, then $\boldsymbol{A}$ is not diagonalizable.

3. Let $S_1 \cup S_2 \cup \cdots \cup S_k = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ with $\boldsymbol{A}\boldsymbol{v}_i = \mu_i \boldsymbol{v}_i$ for all i. Let $\boldsymbol{P} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, and $\boldsymbol{D} = \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)$. Then

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}.$$

Algorithm for Diagonalization

Let $\boldsymbol{A}$ be a square matrix of order n.

1. Find eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$ by solving $\det(x\boldsymbol{I} - \boldsymbol{A}) = 0$.

   If $\det(x\boldsymbol{I} - \boldsymbol{A})$ cannot be completely factorized in $\mathbb{R}$, then $\boldsymbol{A}$ is not diagonalizable.

2. For each (real) eigenvalue λ_i, find a basis S_i for the eigenspace E_{λ_i}.

   If $|S_i| <$ algebraic multiplicity of λ_i in $\det(\lambda\boldsymbol{I} - \boldsymbol{A})$, then $\boldsymbol{A}$ is not diagonalizable.

3. Let $S_1 \cup S_2 \cup \cdots \cup S_k = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ with $\boldsymbol{A}\boldsymbol{v}_i = \mu_i \boldsymbol{v}_i$ for all i.
   Let $\boldsymbol{P} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, and $\boldsymbol{D} = \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)$. Then

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}.$$

Algorithm for Diagonalization

Let $\boldsymbol{A}$ be a square matrix of order n.

1. Find eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$ by solving $\det(x\boldsymbol{I} - \boldsymbol{A}) = 0$.

   If $\det(x\boldsymbol{I} - \boldsymbol{A})$ cannot be completely factorized in $\mathbb{R}$, then $\boldsymbol{A}$ is not diagonalizable.

2. For each (real) eigenvalue λ_i, find a basis S_i for the eigenspace E_{λ_i}.

   If $|S_i| <$ algebraic multiplicity of λ_i in $\det(\lambda\boldsymbol{I} - \boldsymbol{A})$, then $\boldsymbol{A}$ is not diagonalizable.

3. Let $S_1 \cup S_2 \cup \cdots \cup S_k = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ with $\boldsymbol{A}\boldsymbol{v}_i = \mu_i \boldsymbol{v}_i$ for all i.
   Let $\boldsymbol{P} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, and $\boldsymbol{D} = \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)$. Then

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}.$$

Algorithm for Diagonalization

Let $\boldsymbol{A}$ be a square matrix of order n.

1. Find eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$ by solving $\det(x\boldsymbol{I} - \boldsymbol{A}) = 0$.

   If $\det(x\boldsymbol{I} - \boldsymbol{A})$ cannot be completely factorized in $\mathbb{R}$, then $\boldsymbol{A}$ is not diagonalizable.

2. For each (real) eigenvalue λ_i, find a basis S_i for the eigenspace E_{λ_i}.

   If $|S_i| <$ algebraic multiplicity of λ_i in $\det(\lambda\boldsymbol{I} - \boldsymbol{A})$, then $\boldsymbol{A}$ is not diagonalizable.

3. Let $S_1 \cup S_2 \cup \cdots \cup S_k = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ with $\boldsymbol{A}\boldsymbol{v}_i = \mu_i \boldsymbol{v}_i$ for all i.
   Let $\boldsymbol{P} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, and $\boldsymbol{D} = \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)$. Then

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}.$$

Algorithm for Diagonalization

Let $\boldsymbol{A}$ be a square matrix of order n.

1. Find eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$ by solving $\det(x\boldsymbol{I} - \boldsymbol{A}) = 0$.

   If $\det(x\boldsymbol{I} - \boldsymbol{A})$ cannot be completely factorized in $\mathbb{R}$, then $\boldsymbol{A}$ is not diagonalizable.

2. For each (real) eigenvalue λ_i, find a basis S_i for the eigenspace E_{λ_i}.

   If $|S_i| <$ algebraic multiplicity of λ_i in $\det(\lambda\boldsymbol{I} - \boldsymbol{A})$, then $\boldsymbol{A}$ is not diagonalizable.

3. Let $S_1 \cup S_2 \cup \cdots \cup S_k = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ with $\boldsymbol{A}\boldsymbol{v}_i = \mu_i \boldsymbol{v}_i$ for all i. Let $\boldsymbol{P} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, and $\boldsymbol{D} = \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)$. Then

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}.$$

Question 2

PollEv.com/kaimengtan612

Let $\boldsymbol{A} = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}$, $\boldsymbol{B} = \begin{pmatrix} 12 & 1 & 3 \\ -26 & 3 & -17 \\ -14 & -1 & -5 \end{pmatrix}$ and $\boldsymbol{C} = \begin{pmatrix} -1 & 1 & 1 \\ 1 & 1 & -1 \\ -4 & 2 & 3 \end{pmatrix}$.

Determine if these matrices are diagonalizable, and for each matrix that is diagonalizable, find an invertible matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that that matrix equals $\boldsymbol{PDP}^{-1}$.
(For $\boldsymbol{A}$ and $\boldsymbol{B}$, cf. Question 3 of Lesson 16.)

[Pg 44 of Chapter 6.]

Question 2: Solution

$\det(x\boldsymbol{I}-\boldsymbol{A})=(x+2)^2(x-4)$, $E_{-2}=\text{span}\left\{\begin{pmatrix}1\\1\\0\end{pmatrix},\begin{pmatrix}-1\\0\\1\end{pmatrix}\right\}$, $E_4=\text{span}\left\{\begin{pmatrix}\frac{1}{2}\\\frac{1}{2}\\1\end{pmatrix}\right\}$. Let $\boldsymbol{P}=\begin{pmatrix}1&-1&\frac{1}{2}\\1&0&\frac{1}{2}\\0&1&1\end{pmatrix}$ and $\boldsymbol{D}=\begin{pmatrix}-2&0&0\\0&-2&0\\0&0&4\end{pmatrix}$. Then $\boldsymbol{P}$ is invertible, and $\boldsymbol{A}=\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$.

$\det(x\boldsymbol{I}-\boldsymbol{B})=(x+2)(x-6)^2$, $E_6=\text{span}\left\{\begin{pmatrix}-1\\3\\1\end{pmatrix}\right\}$. Thus $\boldsymbol{B}$ is not diagonalizable since the algebraic multiplicity of the eigenvalue 6 is not equal to its geometric multiplicity.

$\det(x\boldsymbol{I}-\boldsymbol{C})=(x^2+2x+2)(x-1)$. Since $x^2+2x+2=(x+1)^2+1\geq 1$ for all $x\in\mathbb{R}$, $\boldsymbol{C}$ is not diagonalizable.

Question 2: Solution

$\det(x\boldsymbol{I}-\boldsymbol{A})=(x+2)^2(x-4)$, $E_{-2}=\operatorname{span}\left\{\begin{pmatrix}1\\1\\0\end{pmatrix},\begin{pmatrix}-1\\0\\1\end{pmatrix}\right\}$, $E_4=\operatorname{span}\left\{\begin{pmatrix}\frac{1}{2}\\\frac{1}{2}\\1\end{pmatrix}\right\}$. Let $\boldsymbol{P}=\begin{pmatrix}1&-1&\frac{1}{2}\\1&0&\frac{1}{2}\\0&1&1\end{pmatrix}$ and $\boldsymbol{D}=\begin{pmatrix}-2&0&0\\0&-2&0\\0&0&4\end{pmatrix}$. Then $\boldsymbol{P}$ is invertible, and $\boldsymbol{A}=\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$.

$\det(x\boldsymbol{I}-\boldsymbol{B})=(x+2)(x-6)^2$, $E_6=\operatorname{span}\left\{\begin{pmatrix}-1\\3\\1\end{pmatrix}\right\}$. Thus $\boldsymbol{B}$ is not diagonalizable since the algebraic multiplicity of the eigenvalue 6 is not equal to its geometric multiplicity.

$\det(x\boldsymbol{I}-\boldsymbol{C})=(x^2+2x+2)(x-1)$. Since $x^2+2x+2=(x+1)^2+1\geq 1$ for all $x\in\mathbb{R}$, $\boldsymbol{C}$ is not diagonalizable.

Question 2: Solution

$\det(x\boldsymbol{I}-\boldsymbol{A})=(x+2)^2(x-4)$, $E_{-2}=\text{span}\left\{\begin{pmatrix}1\\1\\0\end{pmatrix},\begin{pmatrix}-1\\0\\1\end{pmatrix}\right\}$, $E_4=\text{span}\left\{\begin{pmatrix}\frac{1}{2}\\\frac{1}{2}\\1\end{pmatrix}\right\}$. Let $\boldsymbol{P}=\begin{pmatrix}1&-1&\frac{1}{2}\\1&0&\frac{1}{2}\\0&1&1\end{pmatrix}$ and $\boldsymbol{D}=\begin{pmatrix}-2&0&0\\0&-2&0\\0&0&4\end{pmatrix}$. Then $\boldsymbol{P}$ is invertible, and $\boldsymbol{A}=\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$.

$\det(x\boldsymbol{I}-\boldsymbol{B})=(x+2)(x-6)^2$, $E_6=\text{span}\left\{\begin{pmatrix}-1\\3\\1\end{pmatrix}\right\}$. Thus $\boldsymbol{B}$ is not diagonalizable since the algebraic multiplicity of the eigenvalue 6 is not equal to its geometric multiplicity.

$\det(x\boldsymbol{I}-\boldsymbol{C})=(x^2+2x+2)(x-1)$. Since $x^2+2x+2=(x+1)^2+1\geq 1$ for all $x\in\mathbb{R}$, $\boldsymbol{C}$ is not diagonalizable.

Orthogonal Matrix

Let $\boldsymbol{A}$ be a square matrix of order n.

The following statements are equivalent:

1. $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{I}_n$.
2. $\boldsymbol{A}\boldsymbol{A}^T = \boldsymbol{I}_n$.
3. $\boldsymbol{A}$ is invertible, with $\boldsymbol{A}^{-1} = \boldsymbol{A}^T$.
4. The columns of $\boldsymbol{A}$ form an orthonormal basis for $\mathbb{R}^n$.
5. The rows of $\boldsymbol{A}$ form an orthonormal basis for $\mathbb{R}^n$.

When any of the above statements holds, we say that $\boldsymbol{A}$ is an **orthogonal matrix**.

Orthogonal Diagonalization

A square matrix $\boldsymbol{A}$ is **orthogonally diagonalizable** if and only if there exists an orthogonal matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T}.$$

If $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T}$ where $\boldsymbol{D}$ is diagonal, then

$$\boldsymbol{A}^{T} = (\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T})^{T} = (\boldsymbol{P}^{T})^{T}\boldsymbol{D}^{T}\boldsymbol{P}^{T} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T} = \boldsymbol{A};$$

i.e. $\boldsymbol{A}$ is symmetric.

Theorem

If $\boldsymbol{A}$ is symmetric, i.e. $\boldsymbol{A}^{T} = \boldsymbol{A}$, then $\boldsymbol{A}$ is orthogonally diagonalizable.

Corollary

A square matrix $\boldsymbol{A}$ is orthogonally diagonalizable if and only if $\boldsymbol{A}$ is symmetric, i.e. $\boldsymbol{A}^{T} = \boldsymbol{A}$.

Orthogonal Diagonalization

A square matrix $\boldsymbol{A}$ is **orthogonally diagonalizable** if and only if there exists an orthogonal matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T}.$$

If $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T}$ where $\boldsymbol{D}$ is diagonal, then

$$\boldsymbol{A}^{T} = (\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T})^{T} = (\boldsymbol{P}^{T})^{T}\boldsymbol{D}^{T}\boldsymbol{P}^{T} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T} = \boldsymbol{A};$$

i.e. $\boldsymbol{A}$ is symmetric.

Theorem

If $\boldsymbol{A}$ is symmetric, i.e. $\boldsymbol{A}^{T} = \boldsymbol{A}$, then $\boldsymbol{A}$ is orthogonally diagonalizable.

Corollary

A square matrix $\boldsymbol{A}$ is orthogonally diagonalizable if and only if $\boldsymbol{A}$ is symmetric, i.e. $\boldsymbol{A}^{T} = \boldsymbol{A}$.

Orthogonal Diagonalization

A square matrix $\boldsymbol{A}$ is **orthogonally diagonalizable** if and only if there exists an orthogonal matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T}.$$

If $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T}$ where $\boldsymbol{D}$ is diagonal, then

$$\boldsymbol{A}^{T} = (\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T})^{T} = (\boldsymbol{P}^{T})^{T}\boldsymbol{D}^{T}\boldsymbol{P}^{T} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T} = \boldsymbol{A};$$

i.e. $\boldsymbol{A}$ is symmetric.

Theorem

If $\boldsymbol{A}$ is symmetric, i.e. $\boldsymbol{A}^{T} = \boldsymbol{A}$, then $\boldsymbol{A}$ is orthogonally diagonalizable.

Corollary

A square matrix A is orthogonally diagonalizable if and only if A is symmetric, i.e. $A^{T} = A$.

Orthogonal Diagonalization

A square matrix $\boldsymbol{A}$ is **orthogonally diagonalizable** if and only if there exists an orthogonal matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T}.$$

If $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T}$ where $\boldsymbol{D}$ is diagonal, then

$$\boldsymbol{A}^{T} = (\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T})^{T} = (\boldsymbol{P}^{T})^{T}\boldsymbol{D}^{T}\boldsymbol{P}^{T} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T} = \boldsymbol{A};$$

i.e. $\boldsymbol{A}$ is symmetric.

Theorem

If $\boldsymbol{A}$ is symmetric, i.e. $\boldsymbol{A}^{T} = \boldsymbol{A}$, then $\boldsymbol{A}$ is orthogonally diagonalizable.

Corollary

A square matrix $\boldsymbol{A}$ is orthogonally diagonalizable if and only if $\boldsymbol{A}$ is symmetric, i.e. $\boldsymbol{A}^{T} = \boldsymbol{A}$.

Algorithm for Orthogonal Diagonalization

1. Check that the matrix is symmetric; otherwise it is not orthogonally diagonalizable.
2. Same as for diagonalization, except that we need to perform Gram-Schmidt process to convert the basis S_i for E_{λ_i} into an orthonormal basis.

Algorithm for Orthogonal Diagonalization

1. Check that the matrix is symmetric; otherwise it is not orthogonally diagonalizable.
2. Same as for diagonalization, except that we need to perform Gram-Schmidt process to convert the basis S_i for E_{λ_i} into an orthonormal basis.

Question 3

PollEv.com/kaimengtan612

Orthogonally diagonalize

$$\boldsymbol{A} = \begin{pmatrix} 2 & 2 & -2 \\ 2 & -1 & 4 \\ -2 & 4 & -1 \end{pmatrix}.$$

[Characteristic polynomial of $\boldsymbol{A}$ is $(x-3)^2(x+6)$. Pg 57–58 of Chapter 6.]

Question 3: Solution

$$\det(x\boldsymbol{I} - \boldsymbol{A}) = (x-3)^2(x+6).$$

$$3\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} 1 & -2 & 2 \\ -2 & 4 & -4 \\ 2 & -4 & 4 \end{pmatrix} \xrightarrow[R_3 - 2R_1]{R_2 + R_1} \begin{pmatrix} 1 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{ so}$$

$$E_3 = \operatorname{span}\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \right\} = \operatorname{span}\left\{ \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{pmatrix}, \begin{pmatrix} -\frac{2}{\sqrt{45}} \\ \frac{4}{\sqrt{45}} \\ \frac{5}{\sqrt{45}} \end{pmatrix} \right\}.$$

$$-6\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} 8 & -2 & 2 \\ -2 & -5 & -4 \\ 2 & -4 & -5 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \begin{pmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{ so}$$

$$E_{-6} = \operatorname{span}\left\{ \begin{pmatrix} \frac{1}{2} \\ -1 \\ 1 \end{pmatrix} \right\} = \operatorname{span}\left\{ \begin{pmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \end{pmatrix} \right\}.$$

Let $\boldsymbol{P} = \begin{pmatrix} \frac{2}{\sqrt{5}} & -\frac{2}{\sqrt{45}} & \frac{1}{3} \\ \frac{1}{\sqrt{5}} & \frac{4}{\sqrt{45}} & -\frac{2}{3} \\ 0 & \frac{5}{\sqrt{45}} & \frac{2}{3} \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -6 \end{pmatrix}$. Then $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^T$.

Question 3: Solution

$$\det(x\boldsymbol{I} - \boldsymbol{A}) = (x-3)^2(x+6).$$

$$3\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} 1 & -2 & 2 \\ -2 & 4 & -4 \\ 2 & -4 & 4 \end{pmatrix} \xrightarrow[R_3-2R_1]{R_2+R_1} \begin{pmatrix} 1 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{ so}$$

$$E_3 = \operatorname{span}\left\{\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}\right\} = \operatorname{span}\left\{\begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{pmatrix}, \begin{pmatrix} -\frac{2}{\sqrt{45}} \\ \frac{4}{\sqrt{45}} \\ \frac{5}{\sqrt{45}} \end{pmatrix}\right\}.$$

$$-6\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} 8 & -2 & 2 \\ -2 & -5 & -4 \\ 2 & -4 & -5 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \begin{pmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{ so}$$

$$E_{-6} = \operatorname{span}\left\{\begin{pmatrix} \frac{1}{2} \\ -1 \\ 1 \end{pmatrix}\right\} = \operatorname{span}\left\{\begin{pmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \end{pmatrix}\right\}.$$

Let $\boldsymbol{P} = \begin{pmatrix} \frac{2}{\sqrt{5}} & -\frac{2}{\sqrt{45}} & \frac{1}{3} \\ \frac{1}{\sqrt{5}} & \frac{4}{\sqrt{45}} & -\frac{2}{3} \\ 0 & \frac{5}{\sqrt{45}} & \frac{2}{3} \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -6 \end{pmatrix}$. Then $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^T$.

Question 3: Solution

$$\det(x\boldsymbol{I}-\boldsymbol{A})=(x-3)^2(x+6).$$

$$3\boldsymbol{I}-\boldsymbol{A}=\begin{pmatrix}1&-2&2\\-2&4&-4\\2&-4&4\end{pmatrix}\xrightarrow[R_3-2R_1]{R_2+R_1}\begin{pmatrix}1&-2&2\\0&0&0\\0&0&0\end{pmatrix},\text{ so}$$

$$E_3=\operatorname{span}\left\{\begin{pmatrix}2\\1\\0\end{pmatrix},\begin{pmatrix}-2\\0\\1\end{pmatrix}\right\}=\operatorname{span}\left\{\begin{pmatrix}\frac{2}{\sqrt{5}}\\\frac{1}{\sqrt{5}}\\0\end{pmatrix},\begin{pmatrix}-\frac{2}{\sqrt{45}}\\\frac{4}{\sqrt{45}}\\\frac{5}{\sqrt{45}}\end{pmatrix}\right\}.$$

$$-6\boldsymbol{I}-\boldsymbol{A}=\begin{pmatrix}8&-2&2\\-2&-5&-4\\2&-4&-5\end{pmatrix}\xrightarrow[\text{Jordan}]{\text{Gauss-}}\begin{pmatrix}1&0&-\frac{1}{2}\\0&1&1\\0&0&0\end{pmatrix},\text{ so}$$

$$E_{-6}=\operatorname{span}\left\{\begin{pmatrix}\frac{1}{2}\\-1\\1\end{pmatrix}\right\}=\operatorname{span}\left\{\begin{pmatrix}\frac{1}{3}\\-\frac{2}{3}\\\frac{2}{3}\end{pmatrix}\right\}.$$

Let $\boldsymbol{P}=\begin{pmatrix}\frac{2}{\sqrt{5}}&-\frac{2}{\sqrt{45}}&\frac{1}{3}\\\frac{1}{\sqrt{5}}&\frac{4}{\sqrt{45}}&-\frac{2}{3}\\0&\frac{5}{\sqrt{45}}&\frac{2}{3}\end{pmatrix}$ and $\boldsymbol{D}=\begin{pmatrix}3&0&0\\0&3&0\\0&0&-6\end{pmatrix}$. Then $\boldsymbol{A}=\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^T$.

Question 3: Solution

$$\det(x\boldsymbol{I} - \boldsymbol{A}) = (x-3)^2(x+6).$$

$$3\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} 1 & -2 & 2 \\ -2 & 4 & -4 \\ 2 & -4 & 4 \end{pmatrix} \xrightarrow[R_3-2R_1]{R_2+R_1} \begin{pmatrix} 1 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{ so}$$

$$E_3 = \operatorname{span}\left\{\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}\right\} = \operatorname{span}\left\{\begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{pmatrix}, \begin{pmatrix} -\frac{2}{\sqrt{45}} \\ \frac{4}{\sqrt{45}} \\ \frac{5}{\sqrt{45}} \end{pmatrix}\right\}.$$

$$-6\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} 8 & -2 & 2 \\ -2 & -5 & -4 \\ 2 & -4 & -5 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \begin{pmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{ so}$$

$$E_{-6} = \operatorname{span}\left\{\begin{pmatrix} \frac{1}{2} \\ -1 \\ 1 \end{pmatrix}\right\} = \operatorname{span}\left\{\begin{pmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \end{pmatrix}\right\}.$$

Let $\boldsymbol{P} = \begin{pmatrix} \frac{2}{\sqrt{5}} & -\frac{2}{\sqrt{45}} & \frac{1}{3} \\ \frac{1}{\sqrt{5}} & \frac{4}{\sqrt{45}} & -\frac{2}{3} \\ 0 & \frac{5}{\sqrt{45}} & \frac{2}{3} \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -6 \end{pmatrix}$. Then $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^T$.

Next Lesson ...

We will explore **Sections 6.4 and 6.5**.
Please watch/review the video clips for these sections before then.